CROP YIELD PREDICTION USING MACHINE LEARNING

A Regression-Based Approach

Submitted by:

Nikhitha K

Ashritha G

INTRODUCTION

- Agriculture plays a vital role in economic growth and food security
- Crop yield is strongly influenced by climate and soil conditions
- Climate change has increased uncertainty in agricultural productivity
- This project uses machine learning to predict crop yield accurately

OBJECTIVES

- To develop a machine learning regression model
- To predict crop yield using agricultural data
- To analyze the impact of crop and seasonal factors
- To support sustainable agricultural decision-making
- To preprocess and clean agricultural datasets to improve model accuracy
- To evaluate model performance using appropriate regression metrics (R^2, MAE, MSE)

DATA SET DISCRIPTION

Crop-wise and district-wise agricultural dataset

- Features include:
  - State
  - District
  - Crop
  - Season
- Target variable:
  - Crop Yield
- Dataset was processed using Google Colab

DATA PROCESSING

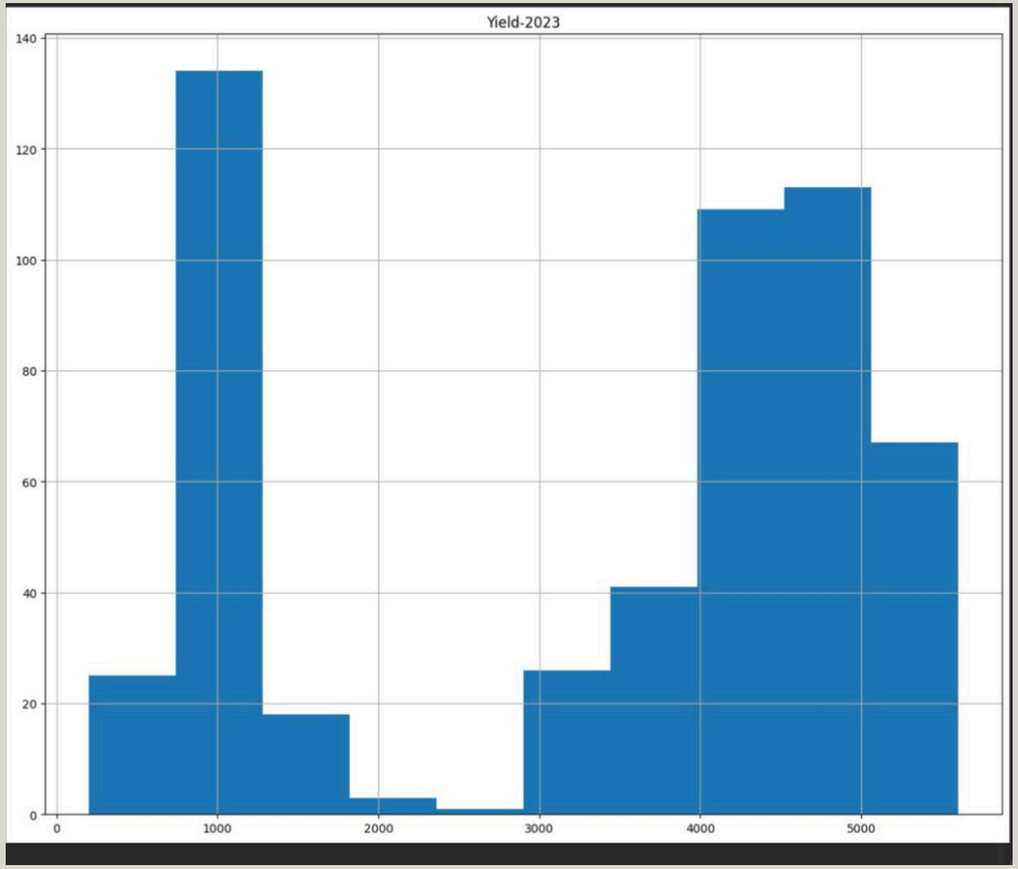
*Removed missing values

* Fixed column naming issues

* Converted categorical data to numerical form

* Prepared data for model training

HISTOGRAM



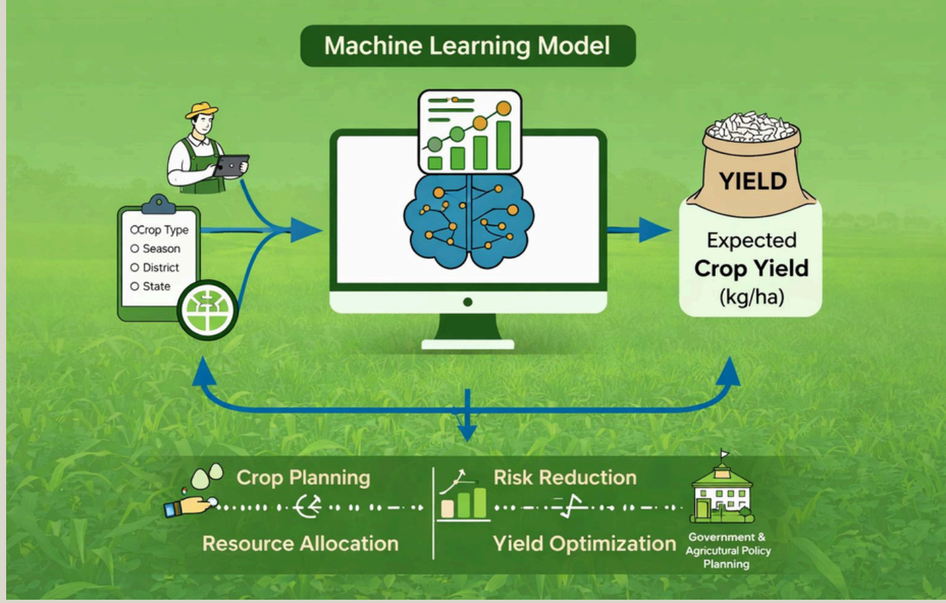
1. The histogram shows the distribution of yield values for 2023, with most observations falling into two separate clusters: one around 800–1500 and another around 3800–5200 units

2. The left cluster (roughly 800–1500) has a tall bar, indicating a high frequency of relatively low yields compared with other ranges.

3. There is a noticeable gap between about 2000 and 3000 on the x-axis, meaning there are very few or no observations with medium yield values in that range.

4. The right cluster (around 3800–5200) is wide with several tall bars, suggesting many observations with high yields and greater variability within this high-yield range.

MODEL APPLICATION



Crop yield prediction is a regression-based machine learning approach that estimates the expected agricultural output using historical data.

The model learns patterns from past records such as crop type, season, district, and state, and establishes a mathematical relationship between these input variables and the target variable (crop yield).

By applying machine learning algorithms, the model generates yield predictions for unseen data, making the process data-driven and objective rather than relying on traditional estimation methods.

Where it is used?

- Farmers – to select suitable crops and estimate expected yield before cultivation
- Agricultural Officers – for monitoring crop productivity across regions
- Government Agencies – for agricultural planning, food security, and policy formulation
- Agri-based Industries – for supply chain and storage planning

CONCLUSION

- Machine learning–based crop yield prediction enhances data-driven agricultural planning and decision-making.
- The proposed model demonstrates how historical agricultural data can be effectively utilized for accurate yield estimation.
- Crop yield prediction contributes to sustainable farming practices by supporting optimal resource utilization.
- The study highlights the practical applicability of machine learning in addressing real-world agricultural challenges

Thank You.

APENDIX

TRAINING AND TESTING

```
[18]
X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.2, random_state=42
        )
```

XG BOOST

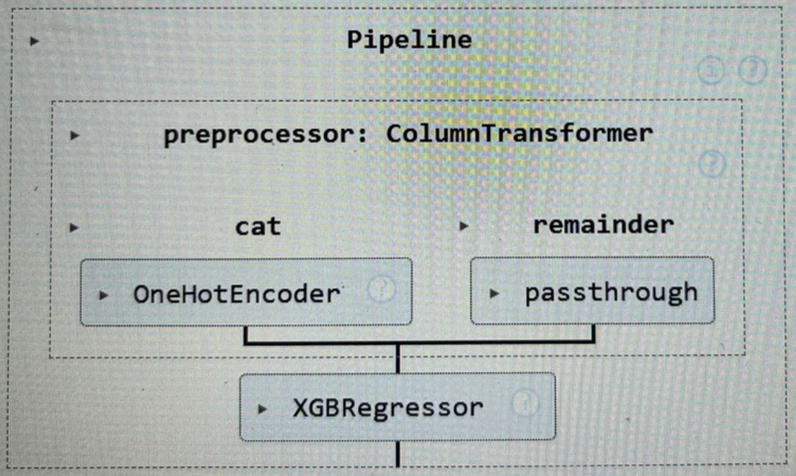
[21]
✓ 0s

```
# Evaluate the model
mse_processed = mean_squared_error(y_test, y_pred_processed)
r2_processed = r2_score(y_test, y_pred_processed)

print("XGBoost Regression Results with Processed Data")
print(f"Mean Squared Error (MSE): {mse_processed}")
print(f"R² Score: {r2_processed}")
```

```
XGBoost Regression Results with Processed Data
Mean Squared Error (MSE): 103865.1015625
R² Score: 0.9689872860908508
```

PIPE LINE

```
model = Pipeline(steps=[
        ('preprocessor', preprocessor),
            ('regressor', xgb_model)
            ])

model.fit(X_train, y_train)
```



TEAM

NIKHITHA.K
ASHRITHA.G